

ABOUBAKR IBRAHIM

Biomedical Engineer | MATLAB | Biomedical Signal Processing | Healthcare Data & Research

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Open to global relocation within one week • Available for research, data and technical roles

PROFESSIONAL SUMMARY

Biomedical Engineer with hands-on MATLAB experience in healthcare data analysis, machine learning and biomedical signal processing. Developed an early sepsis risk-screening graduation prototype using MIMIC-III ICU data and completed paid project-based contributions involving ECG/HRV analysis, QRS detection, research coding and technical reporting. Combines analytical work with hospital clinical-engineering exposure, medical-device knowledge and a strong commitment to reproducible, ethically presented results. Seeking an entry-level Biomedical Data, MATLAB, Research Assistant, Signal Processing or Junior Healthcare Analytics role.

DATA, RESEARCH & TECHNICAL SKILLS

Programming & Analysis	MATLAB; data preparation; feature engineering; statistical analysis; automated tables and visualisation
Machine Learning	Logistic Regression; Random Forest; deep-learning prototypes; classification metrics; class-imbalance awareness
Biomedical Signals	ECG/HRV; R-peak and QRS detection; filtering; frequency-domain analysis; STFT/ISTFT; CNN experimentation
Healthcare Data	MIMIC-III; de-identified clinical data; Microsoft healthcare data concepts; Azure Health Data Services; FHIR fundamentals

RESEARCH & TECHNICAL EXPERIENCE

Freelance Biomedical Research & MATLAB Developer

Project-based

Paid Research and Technical Projects | Remote / International

- Delivered paid MATLAB coding, biomedical-data analysis, signal-processing and research-documentation support for academic and technical projects.
- Implemented ECG/HRV workflows including filtering, adaptive R-peak detection, RR-interval processing, SDNN, RMSSD, pNN50, spectral features and automated CSV/figure export.
- Contributed MATLAB analysis comparing Pan-Tompkins and wavelet-based QRS detection across MIT-BIH records; project ownership remains credited to the academic client.
- Built research prototypes involving ICU mortality modelling, biomedical audio denoising and healthcare technology/risk analysis, with limitations documented for responsible presentation.
- Maintained original code, data workflows, figures and technical reports while preserving client confidentiality and distinguishing personal work from contributed research.

Biomedical Engineering Intern — Clinical Engineering Department

Aug-Sep 2023

Cairo University Hospitals | Cairo, Egypt

- Completed a 24-day internship across ICU, CCU, emergency, operating-room and imaging environments, gaining direct understanding of clinical workflow and equipment-generated data.
- Supported supervised inspection and troubleshooting activities involving ventilators, patient monitors, defibrillators, anesthesia systems, ultrasound and infusion pumps.
- Observed how sensor behaviour, alarms, calibration status, cables and documentation affect data reliability and patient safety.
- Participated in GE CARESCAPE R860 fault investigation involving airflow and pressure sensors and documented maintenance observations.
- Connected hospital equipment experience with later interests in healthcare data quality, clinical decision support and responsible model development.

FLAGSHIP PROJECT

Early Sepsis Detection in ICU Patients Using Machine Learning

Mar–Jul 2025

Işık University | Istanbul, Türkiye

- Designed and coded a MATLAB-based early sepsis risk-screening graduation prototype using de-identified MIMIC-III ICU data.
- Prepared clinical variables and implemented Logistic Regression with Lasso, Random Forest and deep-learning experiments.
- Used accuracy, precision, recall, F1-score and confusion matrices; documented target-definition and validation limitations and established a leakage-safe V2 roadmap.
- Produced the report, code and workflow independently and proposed an ICU monitoring concept connecting biomedical analytics with earlier risk recognition.

EDUCATION

Bachelor of Science in Biomedical Engineering — English Program

2019–2025

Işık University | Istanbul, Türkiye

Graduated: 30 September 2025

SELECTED DATA, HEALTHCARE & QUALITY TRAINING

- Healthcare Data, Azure Health Data Services and FHIR Fundamentals | Microsoft Learn, 2026
- Core Data Concepts and Relational Data in Azure | Microsoft Learn, 2026
- ISO 13485:2016 Medical Device QMS — Internal Auditor | Işık University
- HEARTS Technical Package for Clinicians and Programme Managers | World Health Organization, 2026
- Medical Device Quality Management and Auditing Basics | Scilife Academy, 2026
- Medical Device Quality Assurance and Electrical Safety | Fluke Biomedical, 2026
- McKinsey.org Forward Program — Problem Solving, Communication and Adaptability | McKinsey.org, 2025

ADDITIONAL RESEARCH PORTFOLIO

ECG/HRV analysis • QRS detection: Pan–Tompkins vs Wavelet • ICU mortality modelling prototype • Audio denoising with CNN/STFT • HTA/FMEA and AI-enabled monitoring research

LANGUAGES

Arabic — Native • English — Fluent • Turkish — Basic, actively improving